

Multifamily Takes Baby Steps to CHP



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While energy prices are at record highs with no relief in sight, building owners and managers of multifamily buildings are investigating ways to reduce energy operating costs for their properties. Besides saving energy, these owners and managers are also concerned about the reliability of the electricity supply. Many remember the chaos of the August 2003 blackout in the Northeast, and more recently the power outage in Queens, New York. Avoiding another widespread disruption of the electricity supply has become a priority for both building owners and utilities. There needs to be a better backup system if the grid proves unreliable.

And despite their customers' efforts to conserve energy, suppliers are still seeing an increased need for power capacity. Electric utilities can choose to invest in creating greater power generation capacity—at a cost and timeline that may prove unrealistic—or to create incentives to encourage the installation of privately owned on-site generation. But it takes a tremendous effort to convince customers to create their own electricity, and to coordinate on-site generation with the existing electric grid.

Combined heat and power (CHP) is the production of electricity and the recovery of thermal energy from a single source, typically a natural gas-driven engine or turbine installed on-site. Benefits could be significant, not only from an energy standpoint—the overall efficiency of CHP (60% and more) is about double the overall efficiency of centralized power plants, which face heat losses

at the plant and transmission losses in delivering power from the plant to the customer—but also from an environmental standpoint; the use of CHP could reduce emissions from fossil fuel-burning equipment. CHP sounds like the ideal answer to high energy costs, reliability problems, and the need for increased power production capacity.



Clinton Hill Apartments, Brooklyn, inaugurated in June 2006 the largest installation of advanced microturbine cogenerating units serving a residential complex. The 13 units will generate 600 kilowatts of electricity and hot water for 7 of the site's 12 buildings.

Furthermore, CHP is not a new, untested idea; it is widely used in commercial and industrial applications. There have also been several successful attempts, by HUD in particular, to spur the use of CHP in the multifamily market by publicizing case studies. Unfortunately, incentives for the use of CHP to date have targeted large industrial and commercial projects.

The Opportunity

Currently, funding entities are starting to investigate the use of CHP in multifamily buildings that offer suitable electric and thermal load patterns for CHP applications (see "CHP in the States"). These buildings operate 24 hours a day, seven days a week, and do so with predictable load patterns. Although CHP applications in multifamily buildings would be

on a smaller scale than CHP applications for large industrial users (30 kW for multifamily compared to 500 kW for multi-megawatt industrial applications), the potential exists for many new installations. While investment prices per kW tend to be higher for the installation of these smaller units, prices may decline as companies develop microcogeneration at

1kW of capacity, using a Sterling engine or fuel cell. An efficient technology combined with market volume would then drive prices down. In the meantime, CHP program implementers like the New York State Energy Research and Development Authority (NYSERDA) offer incentives to building owners who want to invest in the technology.

The Challenge

The real challenge at this time is to raise the awareness of, and provide education to, the multifamily buildings sector, explaining the benefits, but also the limitations, of CHP applications. The use of this technology, even with manufacturers' efforts to simplify installation by offering package systems, requires multidisciplinary skills. These include mechanical, thermal, and electrical engineering skills for designing custom applications.

As a principle of CHP, thermal energy must be recovered along with the on-site generation of electricity. Studies have shown that electric utility prices would beat the cost of operating a fossil fuel-driven electric generator without recovery equipment. CHP programs typically require the system to achieve a minimum of 60% overall energy efficiency over the year. Consequently, a feasible CHP project requires coincidence between recoverable thermal loads and electric loads. Potential thermal loads are domestic hot water that offers a predictable pattern

around the year, central heating, central cooling with the use of an absorption chiller, or a heated pool. The electric baseload generally includes common area lighting and, if the building is master metered for electricity, the total of all apartment electric loads.

In addition, CHP implementation involves a strict development process. This process consists of a series of steps that require coordination among the building manager, the engineers, the electricians, the plumbers, the local electric utility for grid connections, and sometimes the local gas utility, and local permitting entities. Prior to procurement, the client should (1) ensure that the site qualifies for CHP; (2) prescreen to estimate cost-effectiveness; (3) and conduct a feasibility study that involves monitoring, CHP selection, energy savings, and capital costs. While an engineering firm would perform a feasibility study, there are several ways to do the qualification and prescreening. Using guidelines or computer-based tools is one way to qualify and prescreen a building for CHP.

Multicogen Tells You If CHP Is Right

Under contract with NYSERDA, Steven Winter Associates (SWA), my employer, developed Multicogen, a Microsoft Excel-based screening tool specifically designed for multifamily buildings, to assess the potential for CHP. We wanted to develop a screening tool that would lead building owners to pursue CHP feasibility studies as the next logical step. NYSERDA is interested in using this tool to promote CHP to multifamily managers and accelerate the deployment of CHP applications statewide. In its current version, the tool can analyze CHP projects located in New York State.

Multicogen first walks the user through a series of multiple-choice questions. Then the software estimates the size and cost-effectiveness of a CHP unit based on building size, local energy pricing, energy metering configuration, domestic hot water use, and heating and cooling system configuration. Based on this information, the tool pro-

vides an early indication of CHP feasibility. Multicogen will flag a Showstopper (a stop sign is actually displayed on the computer screen) if a screening criterion is not met.

One feature of the tool is its multistage screening capability. It can provide a screening result even with minimal information. Although a thorough screening, including analysis of monthly energy records, is recommended, Multicogen can use the thermal and electric load patterns that are typical for multifamily buildings to perform the initial assessment. There is also the option of uploading report files from multifamily energy conservation software such as TREAT or EA-Quip. This option not only avoids redundancy in entering data, but also establishes CHP screening based on accurate estimates of thermal and electric loads determined by these energy conservation software tools.

CHP in the States

At last count (that is, in about 1999), at least 56,000 MW of CHP electricity was being produced in the United States. This translated into 310 billion kWh, 9% of the total U.S. electricity production. The CHP industry, along with DOE, intends to double the amount of CHP production by 2010. If this goal is realized, in 2010 CHP will account for 14% of total U.S. electric generation capacity. The likelihood that this goal will be reached has been bolstered by a recent growth in state funding. The following states offer strong incentives for CHP projects:

New Jersey

New Jersey has declared that its Clean Energy program will continue to reimburse CHP projects through 2006. New Jersey utilizes a three-tiered funding structure; the type of CHP unit and its efficiency determines the total reimbursement that a project can receive. Reimbursements of 30%–60% of total project costs are currently available at up to \$1 million.

New York

The New York State Energy Research and Development Authority (NYSERDA) offers three levels of incentives for CHP and distributed generation (DG) systems. The program is funded with \$15 million yearly. Feasibility studies can receive up to \$100,000; product development projects can receive up to \$500,000; and demonstrations can receive up to \$1 million.

Oregon

Energy Trust of Oregon, an independent nonprofit organization dedicated to energy efficiency and the development of renewable energy, continues to expand its activities in the CHP market. Currently Energy Trust has a program to fund renewable-powered CHP projects. It also has a program that helps to fund fossil fuel-powered CHP projects provided they cost effectively reduce energy use at the facility, and provided the power generated is used on-site. The utility involved

must also be one of the two companies from which Energy Trust receives funding.

California

California CHP users may qualify for tier 3 incentives based on watts produced. Payment varies from \$0.60 to \$0.80 per watt, based on the size of the unit.

Many other states offer some form of incentive for CHP projects. These states include Ohio, Minnesota, Florida, New Hampshire, Kentucky, and Texas, among others. Still other states, such as Vermont, are working on creating CHP incentives as they learn more about the possibilities of CHP. A full list of state-based incentives has been compiled by the American Council for an Energy Efficient Economy (ACEEE) and can be found at www.aceee.org/pubs/ie032full.pdf.

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The design of Multicogen combines guidelines developed by the Midwest CHP Application Center in 2003 and an algorithm involving the integration of energy loads and usage calculations that are specific to multifamily buildings. For instance, the tool performs a spark spread analysis, which is defined as the cost difference in \$/MMBtu between electricity and natural gas. Energy prices largely determine the effectiveness of CHP applications. A minimum threshold of \$12/MMBtu is commonly used for prescreening CHP. This means that a project would not be cost-effective in areas where utilities offer a spark spread below \$12/MMBtu. Consequently, a Showstopper could be flagged early in the screening process, avoiding the need for entering further data such as monthly energy records, which are often tedious to collect and enter in the tool. We learned in an early stage of testing that the tool could be used as a guide in the decision-making process. Users can actually learn the reasons why the use of CHP is or is not feasible.

At any time, users can access a report sheet that gives him or her the results of the screening analysis. When it is not practicable to use CHP in the building, Multicogen also provides suggestions. For example, if the tool determines that a building is individually metered for electricity, it will suggest switching to a master meter with submetering, in order to make it more feasible to use CHP. Reports display a screening checklist; a range of possible CHP sizes; and estimates of capital costs, first-year energy savings, simple paybacks, and savings-investment ratios (SIRs).

Validation Process

As part of the NYSEERDA project, SWA put Multicogen through a validation process. First the tool was tested on more than 100 low-income multifamily

buildings in New York State to verify its capacity to work on buildings of various sizes and configurations. Based on the results of this test, the tool was modified to study CHP production up to 600 kW, up from the original maximum of 400 kW.

Second, the estimates provided by Multicogen were compared to the results of CHP feasibility studies and demonstration projects chosen from among previously funded NYSEERDA projects. The average annual savings estimate provided by Multicogen was within 9% of the results of these more complex studies. This is rather effective, considering the early stage of screening. However, with Multicogen, capital costs for installing CHP systems were found to be 26% lower, on average, than the capital costs of CHP systems in the previous studies.

As a result of the validation process, cost figures used in the Multicogen database were updated to more closely reflect real costs. Following this initial validation process, and confident that Multicogen is an effective screening tool for building owners who are deciding whether or not to do a CHP feasibility study, SWA developed a beta version of the tool for user testing that started last September. Users were selected from among New York State energy auditors and the owners of several multifamily buildings. If the results of the beta test are satisfactory, we can expect the tool to be available to the public in spring 2007.

Although the current version of Multicogen could be made available as a stand-alone Microsoft Excel tool, we foresee a great opportunity in transferring the tool to a Web-based application and expanding it to other states in the future.



FOR MORE INFORMATION:

To learn more about Multicogen, contact the author.

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